

Lumio Project Documentation

PHP + MySQL Workspace Manager

Project Overview

Lumio is a web application for managing a personal workspace or a small team. The application is built with PHP and MySQL and runs locally using XAMPP. Its concept is similar to productivity and note-taking tools: users can create accounts, log in, create pages and notes, upload files, manage tasks and projects, manage calendar events, and view analytics and notifications.

The project uses a user-based data model. Most user-owned data is connected through `user_id` so that each user can access and manage only their own data.

Project Structure

```
project_nti/  
├── index.php  
├── README.md  
├── docs.md  
├── erd_database.png  
├── css/  
├── js/  
├── pages/  
├── sql/  
└── uploads/
```

index.php

The entry point of the project. If the user is logged in, they are redirected to `pages/dashboard.php`. Otherwise, they are redirected to `pages/login.php`.

pages/

Contains the main application pages and logic, including login, registration, dashboard, pages, tasks, files, settings, and other modules.

css/

Contains styling files. There is a general `base.css` file and separate files for modules such as `dashboard.css`, `page.css`, `file_manager.css`, and `calendar.css`.

js/

Contains `script.js`, which mainly improves the page editor by automatically resizing the textarea and displaying the save status.

sql/

Contains lumio.sql, the SQL file responsible for creating the database and tables.

uploads/

Stores files uploaded by users. File metadata is stored in the files table.

Core Shared Files

pages/config.php

Responsible for connecting to the database using PDO and starting the Session if it has not already started.

pages/auth_check.php

Protects restricted pages. If `$_SESSION["user_id"]` does not exist, the user is redirected to the login page.

pages/theme.php

Contains `getUserTheme($pdo, $userId)`, which retrieves the user's theme from `user_settings`. If the theme is missing or invalid, it returns dark.

pages/sidebar.php

The shared sidebar used across most pages. It displays module links and the user's page tree as nested pages through `renderPages`.

User System

Registration — pages/register.php

- Username is required.
- Username must be 3–20 characters.
- Username may contain letters, numbers, and underscores only.
- Email must be valid and unique.
- Password must be at least 8 characters.
- Password must contain an uppercase letter, lowercase letter, number, and special character.
- Password must not contain spaces.

After successful registration, the password is securely hashed using `password_hash` and the user is automatically logged in.

Login — pages/login.php

The application searches for the user by email and verifies the password using `password_verify`. On success, `user_id` and `username` are stored in the Session.

Logout — <pages/logout.php>

Ends the Session and redirects the user to the login page.

Dashboard

File: <pages/dashboard.php>

- Total number of pages.
- Number of root pages without parent_id.
- Number of sub-pages.
- The last five modified pages.

Pages and Notes System

Pages are the core of the project. Each page has a title, icon, content, and can have a parent page through parent_id.

View and Edit Page — <pages/page.php>

Opens a specific page using page.php?id=PAGE_ID. It provides inputs for the icon and title, a content textarea, a save button, a sub-page creation button, and a delete button. The application verifies that the page belongs to the current user.

Create Page — pages/create_page.php

Receives POST data. If parent_id is empty, it creates a root page; otherwise, it creates a sub-page under the selected page. Default title: Untitled Page. Content is empty. After creation, the user is redirected to the new page.

Update Page — pages/update_page.php

Receives id, title, icon, and content and updates the page only if it belongs to the current user. If the title is empty, Untitled Page is used.

Delete Page — pages/delete_page.php

Deletes the page directly from the pages table. Because of ON DELETE CASCADE, deleting a parent page also deletes its child pages. The current trash.php page handles deleted files only, not deleted pages.

File Manager

Main files: pages/file_manager.php, pages/upload_file.php, pages/download_file.php, pages/delete_file.php, <pages/trash.php>

file_manager.php

Displays the user's non-deleted files and provides a file upload form. It can also preview a template when the user comes from the Templates page.

upload_file.php

Uploads a new file to uploads/ and stores its metadata in the files table. Maximum size: 10MB. Allowed extensions: jpg, jpeg, png, gif, pdf, doc, docx, xls, xlsx, txt, zip. A unique filename is generated using uniqid.

download_file.php

Downloads a file by id after verifying that it belongs to the current user and physically exists inside uploads/.

delete_file.php

Does not permanently delete the file. It performs a soft delete by setting deleted_at = NOW().

trash.php

Displays files with deleted_at set and provides Restore by setting deleted_at = NULL, or Delete Permanently by removing the file from the server and the files table.

Templates

File: pages/templates.php

Provides ready-made templates:

- Meeting Notes
- Project Plan
- Daily Notes
- Documentation

When Use Template is clicked, the user is redirected to file_manager.php?template=TEMPLATE_NAME. The file manager prepares the template content in a textarea, which can then be saved in the templates table.

Tasks

File: pages/tasks.php

- Add a task.
- Edit a task.
- Change task status.
- Delete a task.

Task fields: title, description, status, priority, due_date.

Available statuses: Pending, In Progress, Completed.

Creating a new task also creates a success notification.

Projects

File: pages/projects.php

- Add a project.
- Edit a project.
- Change project status.
- Delete a project.

Statuses: Planning, In Progress, Completed.

Priorities: Low, Medium, High.

Calendar

File: pages/calendar.php

- Add an event.
- Edit an event.
- Delete an event.

Event fields: title, description, event_date, event_time, reminder.

Reminder options: No Reminder, 10 Minutes Before, 30 Minutes Before, 1 Hour Before, 1 Day Before.

Team Members

File: pages/team.php

- Add a team member.
- Edit member data.
- Delete a team member.

Fields: full_name, email, role, department.

Notifications

File: pages/notifications.php

- Mark one notification as read.
- Mark all notifications as read.
- Delete a notification.

Notification types: info, warning, success, error.

A notification is currently created automatically when a new task is added.

Analytics

File: pages/analytics.php

- Total pages.
- Total tasks.
- Completed tasks.
- Total projects.
- Total events.
- Total files.

The page_views table exists in the database, but the current code does not record page visits in it.

Profile

File: pages/profile.php

- Username.
- Email.
- Account creation date.
- The first character of the username as a simple avatar.

Settings

File: pages/settings.php

- Update username and email.
- Change password.
- Choose the theme: dark or light.
- Enable or disable Email Notifications.

If the user has no settings in user_settings, default settings are created.

Database

Database creation file:

```
sql/lumio.sql
```

Database name:

```
lumio_db
```

Table	Purpose
users	User and account data
pages	User pages and notes using a tree structure

tasks	User tasks
events	Calendar events
projects	User projects
team_members	Team members
page_views	Intended for page-view tracking, but not currently used by the code
files	Uploaded file metadata
user_settings	User settings such as theme and notifications
notifications	User notifications
templates	Templates saved by the user
trash	Trash table; currently not used by the page deletion logic

Important Relationships

- Most tables are connected to users.id through user_id.
- The pages table is self-related through parent_id, allowing pages inside pages.
- When a user is deleted, related data is deleted because of ON DELETE CASCADE.
- When a parent page is deleted from pages, its child pages are also deleted because of ON DELETE CASCADE.

Main Usage Flow

1. The user opens index.php.
2. If the user is not logged in, they are redirected to login.php.
3. If they do not have an account, they can register through register.php.
4. After login, the user is redirected to dashboard.php.
5. From the Sidebar, the user can navigate to Dashboard, Profile, Tasks, Calendar, Projects, Team Members, Analytics, File Manager, Settings, Notifications, Templates, and Trash.
6. The user can create root pages or sub-pages from the Sidebar or inside an existing page.

Current Code Notes

- Some icons appear incorrectly inside files, such as ðŸ“„. This is likely an encoding issue; saving files as UTF-8 should resolve it.
- trash.php handles deleted files only, not pages.
- The trash table exists in the database, but no current code uses it.
- The page_views table exists, but no current code records page views.
- Most operations use POST and Prepared Statements, which helps reduce SQL Injection risk.
- The application uses htmlspecialchars in many places when displaying user data to reduce XSS issues.